

PROJECT HELIX

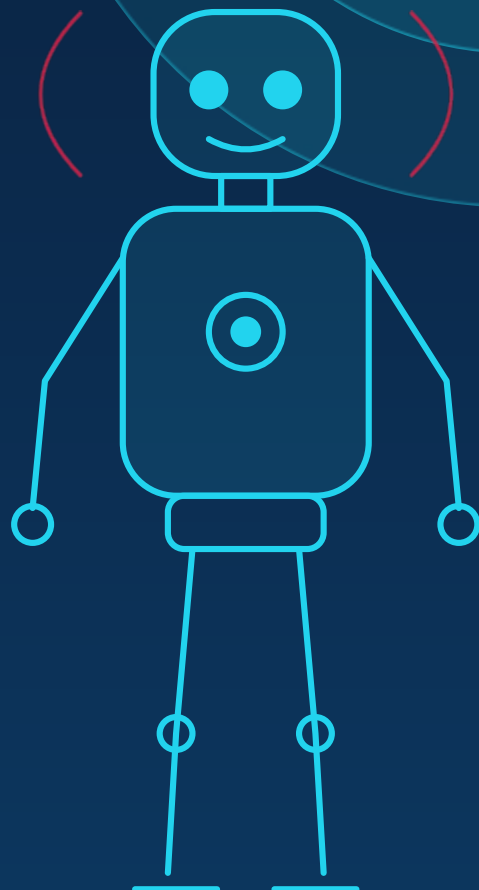
A humanoid assistant for everyday life: it sees, listens, plans and acts safely.

PERCEPTION

LANGUAGE

MOTION

SAFETY



A humanoid that lives alongside people

Millions of people — older adults living alone, people with mobility challenges, busy families — need small, constant help: a reminder to take medication, an object fetched from another room, someone noticing when a fall happens. **Project HELIX** builds a humanoid assistant that perceives its surroundings, understands natural speech (Arabic and English), plans safe actions, and moves with a human-friendly presence.

The project is **simulation-first**: every capability is designed, stress-tested and documented in a physics simulator of the Unitree G1 before any hardware is risked. The result is a fully reproducible software stack that transfers to the real robot.

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DEGREES OF FREEDOM

500 Hz

PHYSICS LOOP

4

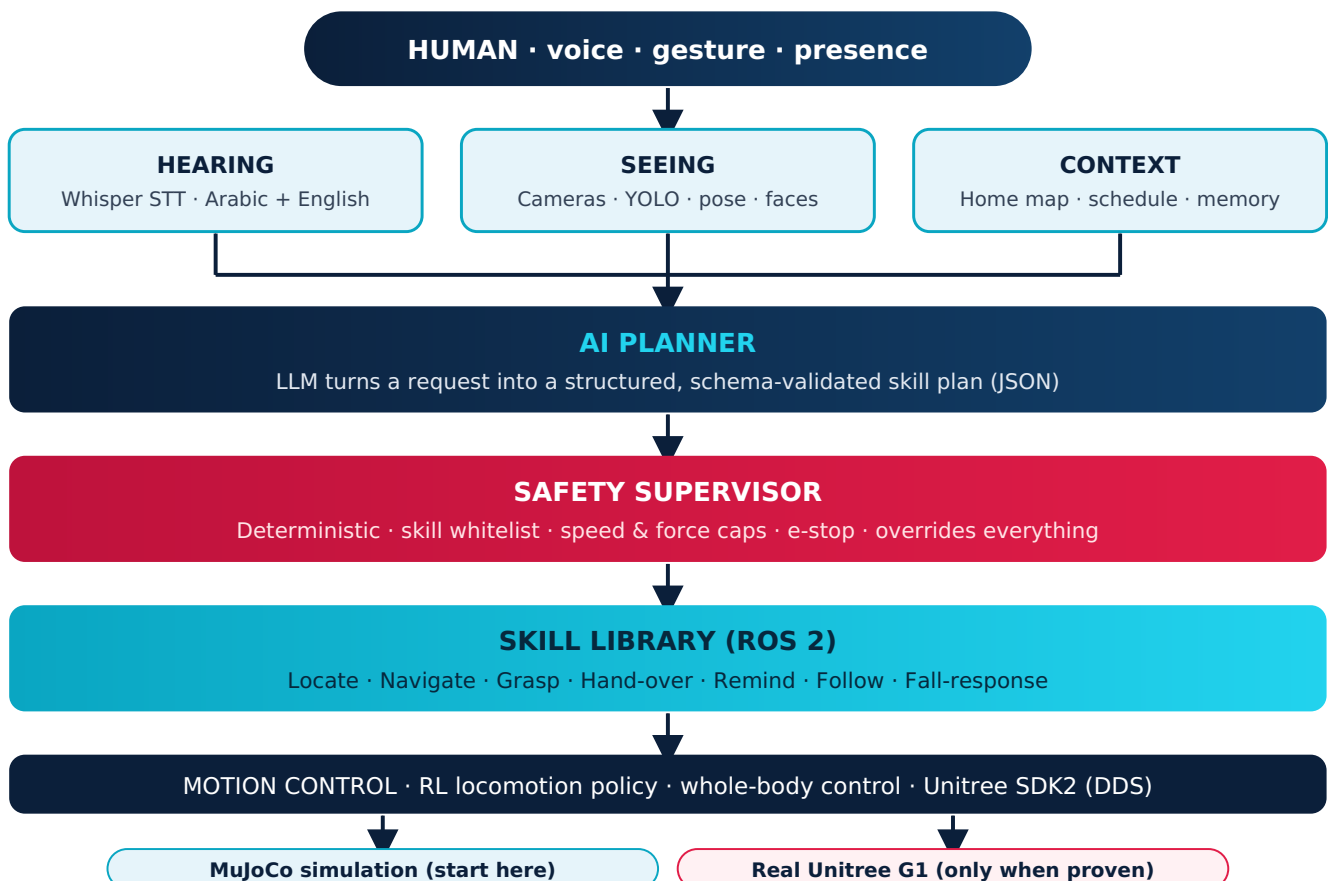
CORE AI LAYERS

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HARDWARE RISK AT START

How the system thinks and acts

Intelligence flows top-down; safety flows bottom-up. A **deterministic Safety Supervisor** sits between the AI planner and the body, so no language-model output can ever move the robot unchecked.



The four intelligence layers

👁 Perception

- Object & person detection (YOLO family)
- Face recognition to greet known people
- Body-pose estimation for posture tracking
- Depth + SLAM for mapping and obstacle avoidance

💬 Language

- Streaming speech-to-text (Whisper, multilingual)
- Natural commands: *"bring me some water"*
- Text-to-speech with a calm, clear voice
- Dialect-aware prompts for Arabic

🧠 Planning

- LLM outputs a structured plan, never raw motor commands
- Plans validated against a JSON schema
- Only whitelisted skills can be called
- Ambiguity triggers a spoken clarifying question

🦾 Motion

- Reinforcement-learned walking policies
- Arm IK for reaching and hand-over
- Train in sim, verify sim-to-sim, then deploy
- Compliant, slow motion near people

From sentence to safe action

The planner converts speech into a plan the supervisor can inspect line by line:

```
// user: "Bring me my water, please"
{
  "intent": "fetch_object",
  "steps": [
    { "skill": "locate",    "target": "cup" },
    { "skill": "navigate", "to": "kitchen", "max_speed_mps": 0.4 },
    { "skill": "grasp",    "object": "cup", "max_force_n": 15 },
    { "skill": "navigate", "to": "user", "max_speed_mps": 0.3 },
    { "skill": "hand_over", "to": "user", "mode": "slow" }
  ]
}
```

Fall detection & response

Signal: body-pose stream shows a rapid torso-angle change, a low head height, then no movement for several seconds.

Response: approach at safe speed → ask aloud “Are you okay?” → if no answer, alert a caregiver with the last camera frame and timestamp.

Trust is engineered, not assumed

- Rule 1 — The AI proposes, the supervisor disposes.** Every plan step passes through a rule-based gate that checks the skill whitelist, speed, force and workspace limits.
- Rule 2 — One button stops everything.** A hardware/software e-stop path is tested before any other feature and always outranks the planner.
- Rule 3 — Fail in simulation first.** Fault injection (sensor dropouts, slips, blocked paths) is run thousands of times in MuJoCo before hardware ever moves.
- Rule 4 — Soft by default.** Reduced speed near humans, force-limited grippers, and a compliant outer layer designed to protect people on contact.

A human-friendly presence

A friendly, lifelike appearance helps people feel at ease, but it must never compromise function. The outer shell is designed as removable, segmented modules on top of the protected robot frame.

LAYER	DESIGN
Skin	Platinum-cure medical silicone, 2–4 mm, cast in segments (forearm, upper arm, torso, thigh, shin, hands), thinner at joints to survive repeated motion.
Under-skin	5–15 mm silicone foam or gel for softness and reduced weight; hidden joint seams; magnetic or zip attachment for quick removal.
Thermal	Concealed vents and airflow paths so motors and battery never overheat under the shell.
Face	Micro-servos for eyelids, brows and mouth; camera eyes; speech-synchronised lip motion; deliberately warm, non-uncanny expressions.
Sizing	The G1 is child-sized (about 127 cm, 35 kg): the shell is proportioned to it rather than faking an adult, and weight is budgeted against joint torque.

Simulation quick-start

Ubuntu or WSL2 on Windows, Python 3. The G1 runs in simulation on your own laptop — no robot required.

```
# 1. environment
python3 -m venv glenv && source glenv/bin/activate
pip install mujoco pygame

# 2. sanity check: opens the MuJoCo viewer
python -m mujoco.viewer

# 3. Unitree SDK for Python
git clone https://github.com/unitreerobotics/unitree_sdk2_python.git
cd unitree_sdk2_python && pip install -e .      # if cyclonedds errors: pip install cyclonedds first

# 4. the G1 simulator
git clone https://github.com/unitreerobotics/unitree_mujoco.git
cd unitree_mujoco/simulate_python
# edit config.py -> ROBOT = "g1"    INTERFACE = "lo"
python3 unitree_mujoco.py
```

Your control code talks to the simulator exactly as it would talk to the real robot, so everything you write in simulation carries over to hardware.

The build roadmap

0

WEEKS 1-2

Foundations

- ☐ Set up Ubuntu / WSL2, Python environment and MuJoCo
- ☐ Launch the MuJoCo viewer and load a robot model
- ☐ Run the Unitree G1 simulator locally
- ☐ Create the GitHub repository with a clear README

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WEEKS 3-5

Body control

- ☐ Read joint states (position, velocity, torque) and log them to CSV
- ☐ Command a single joint with PD control, slowly and safely
- ☐ Move an arm through a sequence of waypoints
- ☐ Add soft joint limits and a velocity clamp
- ☐ Install ROS 2 and bridge SDK topics into ROS 2 nodes

2

WEEKS 6-8

Perception

- ☐ Stream the simulated head camera into a ROS 2 topic
- ☐ Detect people and objects with YOLO
- ☐ Recognise enrolled faces and greet them by name
- ☐ Estimate body pose and implement the fall-detection rule
- ☐ Evaluate detection accuracy on recorded test clips

3

WEEKS 9-11

Language & behaviour

- ☐ Add Whisper speech-to-text (Arabic and English) and text-to-speech
- ☐ Build the LLM planner that outputs schema-validated JSON plans
- ☐ Implement four skills: locate, navigate, remind, hand-over
- ☐ Add spoken clarification when a request is ambiguous

4

WEEKS 12-13

Safety & reliability

- ☐ Write the Safety Supervisor node with whitelist, speed and force caps
- ☐ Implement and test the emergency stop path
- ☐ Run fault-injection scenarios (sensor loss, blocked path, slip)
- ☐ Add full logging and replay for every plan and decision

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WEEKS 14-17 · ADVANCED

Learned locomotion

- ☐ Train a velocity-tracking walking policy with unitree_rl_mjlab or unitree_rl_lab
- ☐ Validate sim-to-sim in MuJoCo under pushes and uneven terrain
- ☐ Compare the learned policy against the built-in controller

6

WEEKS 18-20

Embodiment & hardware readiness

- ☐ Cast one silicone prototype (hand or forearm) and test durability
- ☐ Measure shell weight, heat build-up and joint range impact
- ☐ Confirm the exact G1 variant and its SDK/development access with Unitree
- ☐ First hardware bring-up: robot suspended, shell off, e-stop in hand

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WEEKS 21-22

Showcase

- ☐ Record the demo: medication reminder, fetch-a-cup, and fall response
- ☐ Write the technical report with architecture, results and failure analysis
- ☐ Publish the code, documentation and demo video

08 · TECHNOLOGY STACK

Tools of the build

DOMAIN	TECHNOLOGIES
Languages	Python for AI and rapid iteration; C++ for real-time components
Robot middleware	ROS 2, Unitree SDK2 (DDS), unitree_sdk2_python
Simulation	MuJoCo, unitree_mujoco, Gazebo; Isaac Sim / Isaac Lab for large-scale RL
Learning	PyTorch, reinforcement learning (unitree_rl_mjlab, unitree_rl_lab), imitation learning via LeRobot
Perception & language	OpenCV, YOLO, pose estimation, SLAM; Whisper (STT), a TTS engine, and an LLM planner with JSON-schema validation

Links

RESOURCE	TYPE	LINK
unitree_mujoco — G1 simulator	SIMULATOR	github.com/unitreerobotics/unitree_mujoco
unitree_sdk2_python	SDK	github.com/unitreerobotics/unitree_sdk2_python
G1 MuJoCo simulator (Hugging Face)	SIMULATOR	huggingface.co/lerobot/unitree-g1-mujoco
LeRobot — Unitree G1 guide	GUIDE	huggingface.co/docs/lerobot/unitree_g1
MuJoCo physics engine	ENGINE	github.com/google-deepmind/mujoco
Gazebo	SIMULATOR	gazebo.org
ROS 2 documentation	MIDDLEWARE	docs.ros.org
NVIDIA Isaac Sim (NVIDIA GPU required)	SIMULATOR	developer.nvidia.com/isaac-sim
unitree_rl_lab — RL on IsaacLab	LEARNING	github.com/unitreerobotics/unitree_rl_lab
unitree_rl_mjlab — RL on MuJoCo	LEARNING	github.com/unitreerobotics/unitree_rl_mjlab